Chapter 2: DataLab Foundations

Understanding What You Measured

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1 Overview

This reading explains the cognitive tasks you completed in DataLab and why we collected these data. More importantly, it invites you to reflect on the **practicalities of data collection** — what might have affected the quality and accuracy of your measurements.

> Data Dictionary
>
> For a complete reference of all variables collected, see the Data Dictionary.

2 Reflecting on Data Collection

DataLab was probably your first experience collecting psychological data. You may have noticed things didn't always go smoothly:

- Instructions weren't always clear
- Timing was tricky
- Partners did things differently
- The environment was distracting
- Recording data while running a task was harder than expected

This is normal. In fact, these challenges are exactly why research methods matter. Understanding what can go wrong helps you:

1. Appreciate why standardised procedures exist
2. Interpret your own data more thoughtfully
3. Design better studies in the future

As you read about each task below, think about what happened during *your* data collection. What might have affected your measurements?

3 What You Measured

DataLab included a group memory test plus five tasks measuring different aspects of human cognition and behaviour. For each task, we'll cover the psychological background and then consider methodological issues.

3.1 Task 1: Reaction Time (Reflex Test)

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REFLEX TEST

How fast is your nervous system?

INSTRUCTIONS

1. **Partner A** holds ruler at 30cm mark, hanging vertically
2. **Partner B** positions thumb and finger at 0cm — don't touch the ruler!
3. A drops ruler **without warning**
4. B catches it **as fast as possible**
5. Read the cm mark at the **top of your thumb**
6. Complete **5 trials**, then swap roles

CONVERSION CHART (CM → MILLISECONDS)

cm	1	2	3	4	5	6	7	8	9	10
ms	45	64	78	90	101	111	120	128	135	143
cm	11	12	13	14	15	16	17	18	19	20
ms	150	156	163	169	175	181	186	192	197	202
cm	21	22	23	24	25	26	27	28	29	30
ms	207	212	217	221	226	230	235	239	243	247

Formula: $RT\ (ms) = \sqrt{(2d \div 9.81)} \times 1000$ where d = distance in metres

✍ **RECORD YOUR CATCHES (CM)**

Trial 1	Trial 2	Trial 3	Trial 4	Trial 5

Which hand did you use? ☐ Dominant ☐ Non-dominant

Reaction time is the interval between a stimulus and your response to it. In the ruler-drop task, we measured **simple reaction time** — how quickly you could respond to a single, expected stimulus.

The physics is elegant: when you catch a falling ruler, the distance it fell before you caught it tells us how long your reaction took. Using the equation for free-falling objects ($d = \frac{1}{2}gt^2$), we can convert centimetres into milliseconds.

Why It Matters

Reaction time is one of the most reliable measures in psychology. Your data will show how consistent (or variable) human performance is, even on simple tasks. We can also look for practice effects — did you get faster across trials?

Methodological Considerations

Measurement error:

- Was the ruler held at exactly the same height each time?
- Did the participant's fingers start in exactly the same position?
- Was it always clear where to read the catch point?

Potential biases:

- *Anticipation*: Did you start to predict when the drop would happen?
- *Practice effects*: Did you genuinely get faster, or just better at anticipating?
- *Hand dominance*: We only tested one hand — would results differ for the other?

Procedural issues:

- How consistent was your partner's dropping technique?
- Were there distractions (noise, movement) that affected some trials?
- Did everyone follow the same instructions?

3.2 Memory Test: Mind Palace

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MIND PALACE

Memory test materials

STEP 1: DISTRACTOR TASK (DO THIS FIRST, FOR 60 SECONDS)

Complete as many as you can. This prevents rehearsal of the word list.

47 + 36 =	83 - 29 =	6 × 7 =	54 ÷ 9 =	28 + 45 =
54 + 18 =	91 - 47 =	8 × 9 =	72 ÷ 8 =	63 + 19 =
26 + 57 =	72 - 38 =	4 × 12 =	45 ÷ 5 =	37 + 26 =
63 + 29 =	85 - 56 =	7 × 8 =	63 ÷ 7 =	52 + 31 =
38 + 45 =	64 - 27 =	9 × 6 =	48 ÷ 6 =	44 + 29 =
52 + 39 =	93 - 48 =	5 × 11 =	81 ÷ 9 =	67 + 18 =

STEP 2: WORD RECALL (WRITE ALL THE WORDS YOU REMEMBER)

1	2	3	4	5
6	7	8	9	10

Write words in any order — the numbers are just to help you count.

✍ **TOTAL WORDS RECALLED:** _____ / 10

The memory task measured **working memory** — how many items you can hold in mind while performing another task.

Miller (1956) famously proposed that we can hold about **7 ± 2** items in short-term memory — the "magical number seven." But the distraction task (maths problems) interferes with rehearsal, so performance typically drops.

We'll also examine the **serial position effect**: people tend to remember words from the beginning of a list (primacy) and end of a list (recency) better than the middle. Your position-by-position recall data will let us test this pattern.

⚠ Methodological Considerations

Measurement error:

- Were the words presented at the same speed for everyone?
- Could everyone hear clearly from where they were sitting?
- Was the recall period the same length for all participants?

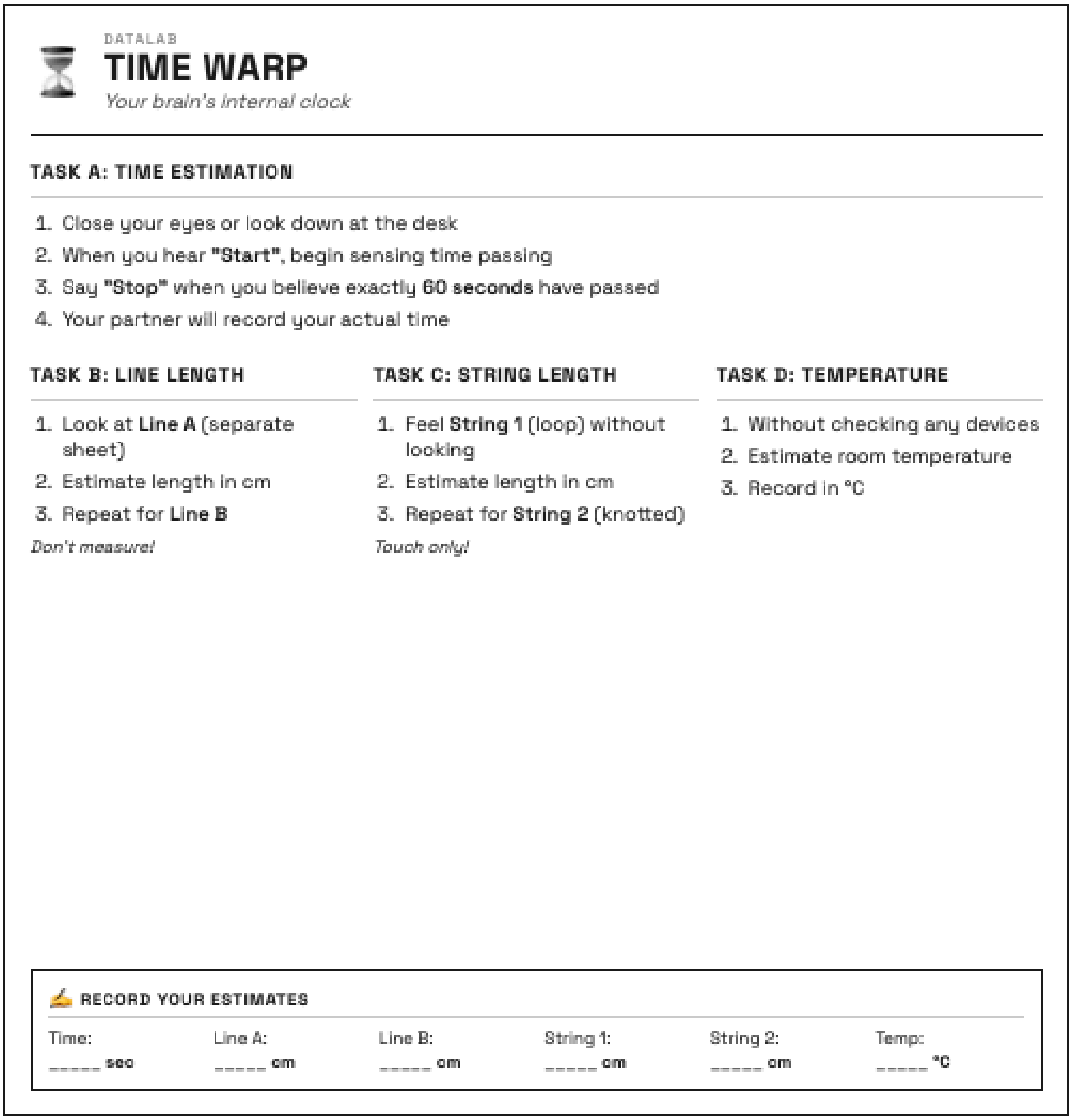
Potential biases:

- *Prior exposure*: Had you seen any of the words recently?
- *Language background*: Were the words equally familiar to everyone?
- *Strategy use*: Some people use chunking or visualisation — should we control for this?

Procedural issues:

- Was the distraction task equally difficult for everyone?
- Did the timing between presentation and recall vary?
- How was recall scored? (exact spelling? close enough?)

3.3 Task 2: Time and Magnitude Perception (Time Warp)



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TIME WARP

Your brain's internal clock

TASK A: TIME ESTIMATION

1. Close your eyes or look down at the desk
2. When you hear **"Start"**, begin sensing time passing
3. Say **"Stop"** when you believe exactly **60 seconds** have passed
4. Your partner will record your actual time

TASK B: LINE LENGTH

1. Look at **Line A** (separate sheet)
2. Estimate length in cm
3. Repeat for **Line B**

Don't measure!

TASK C: STRING LENGTH

1. Feel **String 1** (loop) without looking
2. Estimate length in cm
3. Repeat for **String 2** (knotted)

Touch only!

TASK D: TEMPERATURE

1. Without checking any devices
2. Estimate room temperature
3. Record in °C

✍ **RECORD YOUR ESTIMATES**

Time:	Line A:	Line B:	String 1:	String 2:	Temp:
_____ sec	_____ cm	_____ cm	_____ cm	_____ cm	_____ °C

How accurately do we perceive time? Research shows that our internal clocks are surprisingly inaccurate and can be influenced by many factors.

This station measured three types of estimation:

1. **Time estimation** — How accurately could you judge 60 seconds?
2. **Visual magnitude estimation** — How accurately could you judge line lengths by looking?
3. **Tactile magnitude estimation** — How accurately could you judge lengths by touch alone?

Cross-Modal Comparison

By comparing visual and tactile estimation, we can explore whether some senses are more accurate than others for judging physical properties.

Methodological Considerations

Measurement error:

- When exactly did the stopwatch start? When you said "start" or when your partner pressed it?
- How precisely were line lengths measured?
- Was string tension consistent across tactile trials?

Potential biases:

- *Counting*: Did you count seconds in your head? (This is a strategy, not pure time perception)
- *Anchoring*: Did early estimates influence later ones?
- *Expectation*: Did you adjust estimates based on feedback?

Procedural issues:

- Could you see any clocks or time displays?
- Were there environmental cues (regular sounds, movements)?
- Was it quiet enough to concentrate?

3.4 Task 3: Psychic Staring (Sixth Sense)

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SIXTH SENSE

Can you feel when someone is watching?

INSTRUCTIONS

1. **TARGET** sits with back to partner, eyes closed
2. **STARER** flips a coin: **Heads** = stare at back of partner's head; **Tails** = look away
3. Hold for **15 seconds**, then tap partner's shoulder
4. Target says **"Staring"** or **"Not staring"** and rates confidence
5. Starer reveals the answer; record if correct
6. Complete **6 trials** as target, then swap roles

RECORDING SHEET

Trial	Actual	Your Guess	Confidence	Correct?
1	☐ Staring ☐ Not	☐ Staring ☐ Not	1 2 3 4 5	☐ ✓ ☐ ✕
2	☐ Staring ☐ Not	☐ Staring ☐ Not	1 2 3 4 5	☐ ✓ ☐ ✕
3	☐ Staring ☐ Not	☐ Staring ☐ Not	1 2 3 4 5	☐ ✓ ☐ ✕
4	☐ Staring ☐ Not	☐ Staring ☐ Not	1 2 3 4 5	☐ ✓ ☐ ✕
5	☐ Staring ☐ Not	☐ Staring ☐ Not	1 2 3 4 5	☐ ✓ ☐ ✕
6	☐ Staring ☐ Not	☐ Staring ☐ Not	1 2 3 4 5	☐ ✓ ☐ ✕

Confidence: 1 = pure guess, 5 = absolutely certain

✍ **TOTAL CORRECT: _____ / 6**

Can humans detect when they're being watched? This is a testable empirical question!

The staring detection task uses **signal detection theory**:

- **Hits**: Correctly detecting when someone IS staring
- **False alarms**: Saying "staring" when they're NOT
- **Misses**: Saying "not staring" when they ARE
- **Correct rejections**: Correctly detecting when they're NOT staring

If people cannot actually detect staring, we'd expect performance to be at **chance level** (about 50% correct, or 3 out of 6 trials).

⚠ Methodological Considerations

Measurement error:

- Could you detect your partner through peripheral vision?
- Were there sounds or movements that gave it away?
- Was "staring" defined consistently? (How long? How intense?)

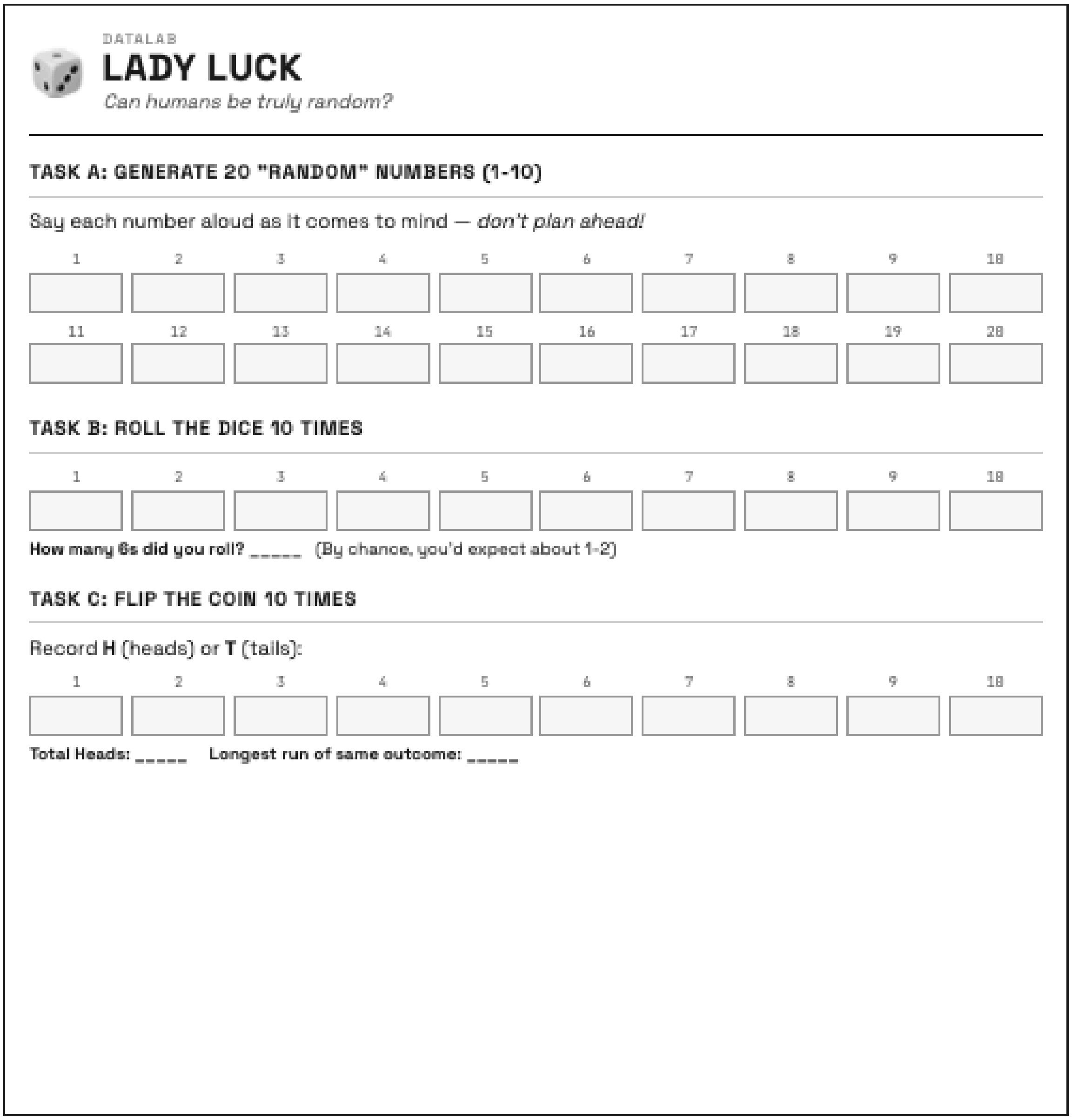
Potential biases:

- *Response bias*: Did you tend to say "yes" or "no" more often?
- *Pattern-seeking*: Did you assume alternating patterns?
- *Confirmation bias*: Did you interpret ambiguous feelings as evidence for your guess?

Procedural issues:

- How reliably did your partner randomise staring/not-staring?
- Was the duration of each trial consistent?
- Could other people in the room affect the "being watched" feeling?

3.5 Task 4: Randomness (Lady Luck)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LADY LUCK

Can humans be truly random?

TASK A: GENERATE 20 "RANDOM" NUMBERS (1-10)

Say each number aloud as it comes to mind — *don't plan ahead!*

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20

TASK B: ROLL THE DICE 10 TIMES

1	2	3	4	5	6	7	8	9	10

How many 6s did you roll? _____ (By chance, you'd expect about 1-2)

TASK C: FLIP THE COIN 10 TIMES

Record **H** (heads) or **T** (tails):

1	2	3	4	5	6	7	8	9	10

Total Heads: _____ **Longest run of same outcome: _____**

Humans are notoriously bad at generating random sequences. When asked to produce "random" numbers, people:

- Avoid repeating the same number twice in a row (even though true randomness includes repetitions)
- Avoid patterns (even though random sequences often contain streaks)
- Favour certain numbers (often 7!)

This station also included dice rolling and coin flipping, which let us compare **true** random outcomes (from dice/coins) with **human-generated** random sequences.

The Gambler's Fallacy

Many people believe that after a streak of heads, tails is "due." This is false — each flip is independent. Your data will likely show longer runs than most people expect!

Methodological Considerations

Measurement error:

- Were all 20 numbers recorded correctly?
- Were dice rolls and coin flips recorded accurately?
- How were "runs" counted? Was this consistent?

Potential biases:

- *Number preferences*: Did cultural factors affect which numbers felt "random"?
- *Avoidance of repetition*: This is a known bias — but did you try to overcome it?
- *Anchoring*: Did your previous number influence your next choice?

Procedural issues:

- Was there time pressure that affected your choices?
- Did you understand what "random" meant in this context?
- Were the dice fair? Were coins flipped properly?

3.6 Task 5: Metacognition (Frontal Lob)

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FRONTAL LOB

Precision meets confidence

INSTRUCTIONS

1. **Before throwing:** Rate your ball-throwing ability (1 = terrible, 10 = excellent)
2. Stand at the marked line
3. Throw ping pong balls at the target
4. Complete **10 throws** with your dominant hand
5. Count how many hit the target
6. *Optional:* Try 5 throws with your non-dominant hand

✍ **RECORD YOUR DATA**

Self-rating:	Dominant hand hits:	Non-dominant hits:	Which hand is dominant?
_____ / 10	_____ / 10	_____ / 5	☐ Left ☐ Right

Metacognition is thinking about your own thinking — including how well you know your own abilities.

The throwing task measured **calibration**: the match between your self-assessed ability and your actual performance.

- **Overconfidence**: Rating yourself higher than your performance
- **Underconfidence**: Rating yourself lower than your performance
- **Well-calibrated**: Your rating matches your performance

Research consistently shows that people are poorly calibrated — often overconfident about easy tasks and underconfident about hard ones.

⚠ Methodological Considerations

Measurement error:

- How were successful throws defined? What about balls that bounced in?
- Who counted the successes — participant or tester?
- Was counting accurate across 10 throws?

Potential biases:

- *Self-serving bias*: Did you rate yourself higher because you wanted to be good?
- *Social desirability*: Did having a partner watching affect your self-rating?
- *Anchoring*: Did the 1-10 scale influence what felt like a "reasonable" rating?

Procedural issues:

- Was the throwing distance the same for everyone?
- Was the target positioned consistently?
- Did throwing technique vary (overarm, underarm)?

4 The Pre-Post Design

DataLab used a **within-subjects** design to measure mood change. You rated your mood, energy, and stress both **before** and **after** the stations.

This design is powerful because each person serves as their own control. By comparing your "before" to your "after," we eliminate individual differences — we're measuring change within you, not differences between people.

! Teaching Application

When we learn about paired t-tests, we'll analyse whether DataLab systematically affected mood, energy, or stress. Did participation make people feel better? Worse? More or less stressed?

5 Why This Matters

You've just experienced something important: **collecting data is hard.**

The methodological issues above aren't criticisms — they're reality. Even professional researchers face these challenges. The difference is that professionals:

1. **Use standardised procedures** — detailed protocols that specify exactly how to do each step

2. **Define variables operationally** — clear definitions of what counts as a "catch" or a "success"
3. **Train testers** — so everyone administers tasks the same way
4. **Control the environment** — quiet rooms, consistent setups, minimal distractions
5. **Check inter-rater reliability** — do two testers recording the same event get the same result?

When we analyse your DataLab data, we'll need to keep these limitations in mind. Some variability in the data reflects genuine individual differences. Some reflects measurement error and procedural inconsistencies.

Learning to tell the difference is what research methods is all about.

6 Why We Generated Our Own Data

6.1 Learning with Real Data

When we analyse data you've generated:

- You understand what the numbers represent
- You can connect statistics to real experiences
- The results are genuinely unknown (not textbook examples)

6.2 Experiencing Both Roles

Acting as both **participant** and **tester** taught you:

- The importance of standardised procedures
- How instructions affect behaviour
- Why research protocols exist

6.3 Understanding Limitations

Having collected data yourself, you now understand:

- Why researchers obsess over procedures
- How small variations can affect results
- That "messy data" is normal — interpretation requires care

7 How You Worked in Pairs

At each task, you worked with a partner:

1. One person was the **participant** (completed the task)
2. One person was the **tester** (administered the task, recorded data)
3. Then you **swapped roles**

This meant everyone experienced both perspectives.

You received a **sequence card** telling you which tasks to complete in which order.

8 Key Terms

Term	Related Concepts
Reaction time	simple RT, catch distance, milliseconds
Working memory	memory span, serial position, primacy/recency
Signal detection	hits, false alarms, sensitivity
Metacognition	self-assessment, calibration
Calibration	overconfidence, underconfidence

9 References

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- Tversky, A., & Kahneman, D. (1974). Judgment under uncertainty: Heuristics and biases. *Science*, 185(4157), 1124-1131.
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